

Yash Soni (Full-time May 26, 2026)

Robotics & Automation Engineer | [Robotics Portfolio](#) | +1(585) 397-6492 | ys8562@rit.edu | www.linkedin.com/in/yash-soni-robotics-engineer | Rochester, NY

PROFESSIONAL EXPERIENCE

Automation Design Engineering Associate

VeroTouch Construction LLC, Salida, CO | January 2026 - Present

- Leading redevelopment of a 3D concrete printing system with ~50% downtime, targeting improved job completion reliability and operational safety.
- Developing a centralized software interface on the **Omron NYB32** controller serving as the system's core for slicer output visualization, monitoring, and diagnostics. Architecting the NYB32 interface as the master communication hub, synchronized with the **NJ501-1400 PLC** for real-time control and system coordination.
- Designing and developing a fully synchronized **mobile pendant interface** with enhanced physical operator controls for safer human intervention and improved command over the robot. Implementing low-latency file conversion and data exchange pipelines to ensure seamless interoperability across software, PLC, and robotic subsystems.
- Reducing downtime and improving safety through unified software, controls integration, and coordinated human-machine interaction.
- Managing task planning and work distribution across team members to support efficient delivery and system integration.

Graduate Research Assistant

Kate Gleason College of Engineering, Rochester, NY | January 2025 - Present

- Developing real-time adaptive control systems for bio-inspired actuators using LabVIEW and MATLAB/Simulink.
- Designed feedback-driven algorithms achieving 10 ms response time, paralleling on-site mechatronic responsiveness.
- Collaborating with interdisciplinary teams to enhance predictive modeling and embedded control reliability.

Software Engineer

Searce Co-Sourcing Pvt. Ltd., India | January 2022 – July 2024

- Built and deployed automation pipelines, improving system reliability by 40%.
 - Applied distributed systems design and infrastructure automation principles for scalable cloud-based solutions, reflecting experience in robust system development for real-world deployment
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TECHNICAL SKILLS

- **Specialized Skills:** Autonomy Algorithms, Motion Planning, Real-Time Control Systems, Estimation & Sensor Fusion, Version Control Systems, Agile Methodologies
 - **Software & Tools:** PyTorch, TensorFlow, Autodesk AutoCAD, Inventor, Fusion360, ROS1/ROS2, SolidWorks, SQL, ABB RobotStudio, Gazebo, RViz, OpenCV, Google Cloud, AWS, LabVIEW, Docker, Spectrum Analyzers, Multimeters, Control Systems Analysis
 - **Microcontrollers & Development Boards:** Arduino, Raspberry Pi, NodeMCU, PLC Programming
 - **Programming Languages:** C++ (Object Oriented Programming, STL, multithreading, optimization), Python (NumPy, SciPy), Java, Go, C, JavaScript (React JS, Node JS, Angular)
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PROJECTS

- **Automated Assembly Line Cell with Vision-Based Quality Inspection (ABB Robot + PLC Integration)**
 - Developed an autonomous decision-making module with **PLC-integrated robotic cell** using an ABB controller and **OpenCV-based vision inspection** for real-time quality correction during continuous production.
 - Configured **digital/analog I/O** and implemented **Ethernet/IP communication** for synchronized conveyor and robotic control.
 - Achieved a **production-ready automation cell** validated through commissioning and cycle testing for real-time defect rejection.
 - Integrated **vision sensors, PLCs, and robot controllers**, and configured **industrial networking** (IP setup, switches, and fieldbus protocols) for seamless operation.
 - **Adaptive Free-form 3D Printing Mechatronic System using ABB CRB 15000 Co-bot for On-Site Automation (2025)**
 - Integrated ABB CRB 15000 cobot with a custom extrusion tool and multi-sensor feedback for adaptive filament deposition.
 - Developed autonomous motion planning and real-time control algorithms in MATLAB/Simulink with ROS2 interfaces, achieving <2% deposition variance. Incorporated Multi-objective Reinforcement learning for live obstacle avoidance, with 95% accuracy in performance.
 - Simulated operations in Gazebo, optimizing path planning and synchronization for live construction settings.
 - Integrated PLC-based HMI for real-time flow monitoring and achieved a 35% improvement in print path precision.
 - **Zone-Following Roverbot using SLAM & Detection and tracking of moving objects (DATMO) (2024 - Ongoing)**
 - Implemented simultaneous localization and mapping (SLAM) and object detection/tracking algorithms in Python and C++.
 - Designed an autonomous rover using SLAM and DATMO for real-time mapping and object tracking at nearly **95%** lower cost.
 - Created a fully functional prototype requiring minimal to no user training, capable of autonomous navigation and object tracking with **>90%** accuracy and voice-based user alerts.
 - **ABU ROBOCON 2021 Arrow throwing and Arrow picking Semi-Autonomous Mobile Robots- All India Rank 16/100**
 - Designed and manufactured a pair of robots to autonomously throw and pick arrows in a competitive match against other teams.
 - Implemented semi-autonomous functionality with PID-based trajectory correction for navigation, and pneumatic actuators for arrow pick-and-throw mechanisms.
 - Spearheaded integration of subsystems in collaboration with a **50-member interdisciplinary team** across Mechanical, Electrical, and Computer Engineering.
 - Achieved **16th rank nationally out of 100+ teams**, demonstrating successful performance under strict constraints.
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EDUCATION

Rochester Institute of Technology, Rochester, NY

Master of Science in Mechanical Engineering with Robotics Focus Area, **Sept 2024 - May 2026: GPA 3.89**

MESWCOE, University of Pune, India

Bachelor of Engineering – Mechanical Engineering, **Aug 2019 - Aug 2022: GPA 3.75**